

What is LDAP Enumeration?

LDAP Enumeration is the process of **gathering information** from an **LDAP (Lightweight Directory Access Protocol)** directory service — such as **Microsoft Active Directory** — without logging in as a privileged user.

Simple Explanation:

Think of **LDAP** like a phonebook that stores info about all users, computers, groups, and more in a company network.

LDAP enumeration means trying to **read that phonebook** to find out who's on the network and how it's structured.

What Can Be Enumerated?

What You Can Find Example

✓ Usernames	john.doe, admin, svc_backup
✓ Groups	Domain Admins, HR, IT
✓ Computers	DESKTOP-01, DC01, LAPTOP123
✓ Emails	john.doe@company.local
✓ Password Policies	Minimum password length, lockout rules
✓ Domain Structure	How users and groups are organized

LDAP Default Port

Protocol	Port	Description
LDAP	389	Unencrypted
LDAPS	636	Encrypted with SSL

Tools for LDAP Enumeration

Tool	Description
ldapsearch	Linux CLI tool to query LDAP servers
nmap	With NSE scripts like ldap-search, ldap-novell-root
Ldapdomaindump	Python tool for extracting AD structure (needs credentials)
ADExplorer	GUI tool by Microsoft to browse AD

Common LDAP Query Example (Using Ldapsearch)

```
ldapsearch -x -h 192.168.1.10 -p 389 -b "dc=codingseekho,dc=local"
```

Option Meaning

- x Use simple authentication
- h IP/host of LDAP server
- p Port (389)
- b Base DN (domain to search in)

✅ This command tries to pull all available entries from the LDAP server.

Base DN Example:

If the domain is corp.local, then base DN would be:

```
dc=corp,dc=local
```

When LDAP Enumeration Works

- LDAP **allows anonymous binds** (logins without password)
- LDAP uses **weak access permissions**
- **SMB and NetBIOS** may help discover the domain info (used with Enum4linux or Nmap scripts)

How to Prevent LDAP Enumeration

Step	Reason
Disable anonymous LDAP access	Blocks unauthenticated queries
Use secure authentication	Avoid simple bind without encryption
Enable auditing of LDAP queries	Detects unusual enumeration attempts
Apply least privilege policies	Restrict who can view directory info

Real-World Use Case in Hacking

A penetration tester may:

1. Use Nmap to find open port 389
2. Try anonymous LDAP enumeration with Ldapsearch
3. Find usernames, groups, and trust relationships
4. Use those for password spraying or privilege escalation